Comparative Evaluation of DAB and CLLC Bidirectional DC-DC Converters for Wide-Voltage-Range V2G Chargers

Thantham Khongsamret
Department of Electrical Engineering
Faculty of Engineering at Sriracha
Kasetsart University, Chonburi, Thailand
thantham.k@ku.th

Wason Tanjaroen
Department of Electrical Engineering
Faculty of Engineering at Sriracha
Kasetsart University, Chonburi, Thailand
wason@eng.src.ku.ac.th

Uthane Supatti
Department of Electrical Engineering
Faculty of Engineering at Sriracha
Kasetsart University, Chonburi, Thailand
uthane@eng.src.ku.ac.th

Abstract—This paper compares predefined single-phase-shift dual-active-bridge (SPS-DAB) and asymmetric CLLC bidirectional dc–dc converters in PSIM. The study covers a 400-V dc link, 250–450-V battery voltage, 1–6-kW power, and both grid-to-vehicle and vehicle-to-grid directions. Sixty operating-point pairs, each containing one record per converter, satisfy common numerical qualification criteria. In the baseline modeled-efficiency map, the CLLC nominally leads 31 pairs (27 of 30 at 1–3 kW), whereas the SPS-DAB nominally leads 29 (26 of 30 at 4–6 kW, 19 of them by less than 0.25 pp). However, 33 orderings would reverse if no more than 10 W of hypothetical differential omitted loss were assigned to the nominal winner. Model-based screening classifies 12 pairs as supported, 44 as warning, and four as requiring additional validation. An ideal SPS reconstruction matches simulated DAB primary rms current within 1.8% at all 60 points. At three interpretable operating points, replacing the CLLC parameterized bridges with explicit MOSFETs reduced included loss by 3.37–13.58 W and reversed one near tie. These local checks do not bound model-form uncertainty across the full operating map. The results support conditional screening of the evaluated implementations, not a universal or hardware-validated topology ranking.

Keywords—bidirectional dc–dc converter; CLLC resonant converter; dual-active bridge; efficiency comparison; electric-vehicle charging; G2V; V2G

I. Introduction

A common bidirectional electric-vehicle charger architecture uses an isolated dc–dc stage that can transfer power between a regulated dc link and a battery over a wide voltage and load range [1], [2]. The dual-active bridge (DAB) offers direct phase-shift power control and a simple bidirectional structure, whereas the CLLC resonant converter uses a frequency-dependent tank to shape current and extend soft-switching operation. Their practical ordering is therefore operating-point dependent rather than intrinsic to the topology name [3]–[5].

Previous studies compare isolated charger topologies at different ratings, devices, magnetic designs, and control objectives [2]–[5]. Related work addresses DAB modulation and SiC implementation [6], [7], resonant-stage design and analysis [8]–[10], and resonant-DAB control [11]. Other studies examine wide-range resonant stages and hybrid control [12]–[16], optimized DAB modulation [17], high-frequency CLLC [18], and dual DAB–CLLC hardware [19]. Building on these trade-offs, this work evaluates a defined 400-V-link, 250–450-V, 1–6-kW bidirectional matrix and quantifies how numerical and loss-model assumptions limit its interpretation.

The contribution is a common 60-pair G2V/V2G comparison of two predefined implementations, with numerical qualification and traceable pair-level evidence states. The study relates modeled efficiency and current stress to voltage ratio, examines exact-power, timestep, omitted-loss, temperature, soft-switching, and bridge-model sensitivities, and checks SPS current and CLLC fundamental phase against analytical approximations. Together, these layers yield conditional operating-region screening rules without claiming optimized-topology superiority.

A. Charger Context, Operating Modes, and Boundary

Fig. 1 places the isolated dc–dc stage under comparison within a two-stage charger. In grid-to-vehicle (G2V) operation, the grid-side converter regulates the dc link, and the dc–dc stage delivers positive received power to the battery. In vehicle-to-grid (V2G) operation, the battery-side stage returns positive received power to the dc link for inversion to the grid. The compared records contain dc-link and battery-side electrical quantities; grid-current power quality, battery state-of-charge dynamics, and protection coordination lie outside the simulated boundary.

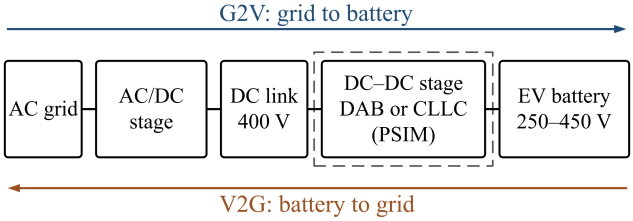


Fig. 1. Two-stage bidirectional charger and model boundary. Only the isolated dc–dc stage (SPS-DAB or asymmetric CLLC) is simulated in PSIM; the ac/dc stage and the battery are system context. Arrows indicate G2V and V2G power flow.

The external matrix fixes the dc link at 400 V and sweeps five battery voltages (250–450 V), six power targets (1–6 kW), and both directions, yielding 60 pairs. This envelope deliberately crosses step-down and step-up conversion, light and high load, and both power-flow polarities.

B. Related Work and Research Gap

DAB–CLLC comparison across different ratings and design objectives is motivated in [2]–[5] and [19]. A broader set of V2G/G2V charger topologies is reviewed in [20], while system power paths for an interleaved nonisolated charger are described in [21]. Table I summarizes how these studies differ from the present comparison. They frame mechanisms and metrics; their numerical results are not imported into the present models.

C. Comparison Criteria and Evidence Types

Switched-model outputs include power, current, waveforms, and gate timing. They are kept separate from derived metrics—modeled loss, modeled efficiency, and

sensitivity classifications—and from literature-based design properties. Only qualified simulation records and the calculations defined in Section II enter the numerical ordering.

TABLE I
PRIOR STUDIES AND EVIDENCE BASIS

Study	Scope	Evidence basis	Distinction from this work
[2]	1-kW isolated bidirectional converters	Analysis and hardware experiments	Different rating and implementations
[3]	3.7-kW GaN DAB and CLLC	Time-domain modeling and hardware	Different devices, frequencies, and design objectives
[4]	10.9-kW on-board charger topologies	Design and charging prototype tests	Different power level and component design
[5]	DAB vs CLLC for EV use	MATLAB/Simulink; SPS	Different parameters and control boundary
[19]	25-kW dual DAB–CLLC; 200–1000 V	Design and experimental validation	Different rating, hardware, and control objectives
This work	DAB vs CLLC; 400-V link; 250–450 V; 1–6 kW; G2V/V2G	60 qualified switched-model pairs plus sensitivity layers	Common external matrix and pair-level evidence state

II. Converter Models and Comparison Methods

A. Converter Implementations and Shared Operating Conditions

Both models share the external operating matrix, the 400-V dc-link boundary, and the qualification rules, while each retains its own modulation. Each result is a settled steady-state operating point. The DAB uses single-phase-shift (SPS) control with both bridges at full duty and controls power with a single phase shift φ at 100 kHz. The asymmetric CLLC uses a fixed schedule: the switching frequency is fixed for each battery voltage and direction (90, 95, 98, or 100 kHz), the duty ratio D of the higher-voltage bridge rises linearly with power from its 1-kW value to unity at 6 kW, and only the receiving-bridge phase is searched to meet the power target. At 1 kW, this schedule equalizes the fundamental amplitudes of the two bridge voltages, V sin(πD/2), after referral through n = 1.1. The CLLC therefore operates with matched fundamental voltages at light load, whereas the SPS-DAB has no equivalent voltage-matching degree of freedom. This is a property of the commanded schedule, not a loss result, so common external conditions do not imply equal optimization; the full 60-point schedule is available from the first author upon reasonable request.

Fig. 2 shows the two evaluated circuits, and Table II lists their parameters. The SPS-DAB uses an 18.333-μH referred transfer inductance, a 3.8236-mH magnetizing inductance, a 1.1:1 transformer ratio, and 200-ns dead time. The asymmetric CLLC uses 30 μH/132 nF and 27.5 μH/144 nF resonant branches, a 130-μH magnetizing inductance, the same ratio and dead time, and the 90/95/98/100-kHz schedule. Equation (1) expresses ideal SPS-DAB power transfer, whereas the simulated primary-rms-current trend with voltage-ratio mismatch is examined separately in Section III-B. The two branch resonances in (2) coincide at 79.98 kHz, although the primary-referred secondary resonant inductance is 33.275 μH versus 30 μH on the primary. Thus, the tank is asymmetric in impedance, and its schedule operates above resonance at f_s/f_r = 1.125–1.250. These are predefined settings, not optimized designs. With 22 turns, the nominal ungapped E65/32/27 N87 AL of 7,900 nH/turn² [22] gives 3.8236 mH; 130 μH instead requires an effective AL of about 269 nH/turn². Gap/fringing performance is not experimentally established.

$$P_{\mathrm{DAB}} = nV_{\mathrm{dc}}V_{\mathrm{bat}}\varphi(\pi - |\varphi|)/(2\pi^2 f_s L_{\mathrm{eq}}),\ -\pi \le \varphi \le \pi \quad (1)$$

$$f_{\mathrm{r1}} = 1/[2\pi\sqrt{(L_{\mathrm{r1}}C_{\mathrm{r1}})}] \approx f_{\mathrm{r2}} \approx 79.98\ \mathrm{kHz} \quad (2)$$

Received power is positive at the destination port: the battery in G2V and the dc link in V2G. The DAB phase and the CLLC receiving-bridge phase are searched against the same power target, while the CLLC frequency and duty schedule remain fixed. No loss or current objective and no outcome-dependent retuning are used. A positive Δη denotes a higher CLLC modeled efficiency.

Both implementations use semiconductor reference [23], PSIM element documentation [24], and a common offline loss-lookup framework wherever applicable. However, the DAB uses discrete MOSFET elements and the CLLC parameterized bridges; identical datasheet references therefore do not establish equal commutation fidelity, switching trajectories, or passive losses. This model-form difference remains a limitation of the comparison.

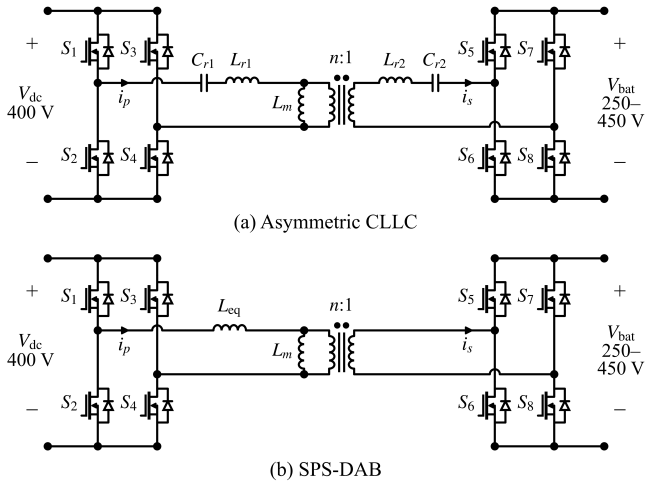


Fig. 2. Evaluated circuits: (a) asymmetric CLLC and (b) SPS-DAB, both with n = 1.1 (22:20) and full-bridge switches S_1–S_8. Component values are listed in Table II.

TABLE II
FIXED MODEL AND CONTROL PARAMETERS

Parameter	SPS-DAB implementation	Asymmetric CLLC implementation
Control and switching frequency	SPS; 100 kHz; signed phase sets direction; $\|\varphi\|$ = 3.978°–51.749°	90/95/98/100 kHz; fixed duty schedule; receiver phase searched only for power error
Reactive network	L_{eq} = 18.333 μH; L_{m} = 3.8236 mH	$L_{\mathrm{r1}}/C_{\mathrm{r1}}$ = 30 μH/132 nF; L_{m} = 130 μH; $L_{\mathrm{r2}}/C_{\mathrm{r2}}$ = 27.5 μH/144 nF; $f_{\mathrm{r}} \approx$ 79.98 kHz
DC link and turns ratio	400 V; $N_{\mathrm{p}}/N_{\mathrm{s}}$ = 1.1	400 V; $N_{\mathrm{p}}/N_{\mathrm{s}}$ = 1.1
Command dead time	200 ns per edge	200 ns per edge
Semiconductor representation	Explicit static MOSFETs; offline C3M0032120K loss data [23]; T_{j} = 125 °C	Parameterized bridges; same offline loss data [23]; T_{j} = 125 °C
Shared external matrix	250–450 V; 1–6 kW; G2V and V2G	250–450 V; 1–6 kW; G2V and V2G

B. Simulation Qualification and Data Selection

Each operating-point pair contains one DAB and one CLLC record. Both must meet identical power, settling, state,

current, waveform, gate, and commutation criteria at a common timestep. The predefined evidence-state rule classifies second-run-only qualified pairs as warnings and the remaining pairs by the sensitivity checks in Section IV. Nominal lead and evidence state are reported separately.

Each record is evaluated on an integer-cycle native-waveform window after staged settling. Qualification requires valid waveform import and gates. Power-target error, state residual, and half-window power drift must be ≤0.5%; two-stage agreement must be ≤0.1% of target power. Modeled loss and primary/secondary rms currents must vary ≤0.1% between stages, and peak currents ≤1%. Half-window $\Delta\eta$ is diagnostic, not an acceptance gate.

Of 60 pairs, 53 qualified in the first 1-ns run, four in a predefined second 1-ns run, and three at bilateral 0.5 ns (G2V 450 V, 3–5 kW). The latter include a CLLC netlist-export failure at 3 kW and gate-pattern failures at 4–5 kW at 1 ns. Both converters were evaluated at the same finer timestep, yielding 57 pairs at 1 ns and three at 0.5 ns. No mixed-timestep comparison or substituted result is used. Independent integration of 120 native windows reproduces reported port power and rms/peak current; netlists and loss records are available from the first author upon reasonable request.

C. Modeled-Loss and Interpretation Boundaries

Modeled efficiency follows (3), where P_{rec} is the received power and $P_{loss,inc}$ is the included loss: MOSFET conduction and switching-energy lookup, transformer copper and core loss, CLLC capacitor equivalent series resistance (ESR), gate charge, and maximum driver quiescent loss. Transfer and resonant inductors are ideal elements; their winding and core losses are excluded. L_m represents the transformer magnetizing branch, so an additional independent core-loss term must not be counted twice. The N87 transformer-loss estimate uses assumed winding geometry and AC resistance; gap/fringing effects are not validated. Controller, auxiliary, busbar, connector, thermal-network, and unparameterized parasitic losses are also excluded. Thus, (3) is a conditional modeled metric, not measured end-to-end efficiency.

$$\eta_m = 100P_{rec}/(P_{rec} + P_{loss,inc}) \quad (3)$$

$$L_{rev} = 100P_{rec,w}/\eta_{m,l} - P_{rec,w} - P_{loss,inc,w} \quad (4)$$

In (4), subscripts w and l denote the nominal winner and loser; $\eta_{m,l}$ is in percent, and $P_{rec,w}$, $P_{loss,inc,w}$, and L_{rev} are in watts. L_{rev} is the differential omitted loss at which the ordering reverses. Soft-switching margins reconstructed from waveforms and capacitance-energy assumptions are diagnostic screens, not measured V_{DS} at turn-on or hardware zero-voltage/zero-current switching (ZVS/ZCS) evidence. The 25, 125, and 175 °C check varies device-model parameters; it is not a thermal-network or hardware temperature result.

III. Simulation Results

A. Numerical Coverage and Validation Status

Table III summarizes the simulation coverage and validation status. All 60 voltage–power–direction pairs have a signed modeled-efficiency difference calculated from two qualified simulation records. Numerical ordering and evidence classification are reported separately. A complete efficiency map does not imply equal confidence at every operating point.

TABLE III
SIMULATION AND VALIDATION SUMMARY

Numerical basis	Outcome	Role in reported comparison
Primary qualified 1-ns simulations	53 pairs	Two qualified records per cell
Predefined second 1-ns run	4 pairs	All CLLC-favored; warning state retained
Predefined 0.5-ns runs (both converters)	3 pairs	G2V 450 V at 3–5 kW
Final numerical coverage	60/60	29 DAB; 31 CLLC
Exact-power assessment	15 of 17 eligible	2 unresolved; nominal values unchanged
Evidence classification	12 / 44 / 4	Supported / warning / additional validation

B. Modeled-Efficiency Trends Across Operating Regions

Fig. 3 reports signed modeled-efficiency differences and Fig. 4 separates nominal leads from omitted-loss sensitivity. Across 60 qualified pairs, CLLC leads 31 and DAB 29, with $\Delta\eta$ ranging from −0.952 to +5.246 percentage points (pp). CLLC leads 9/10, 10/10, and 8/10 at 1–3 kW; DAB leads 7/10, 9/10, and 10/10 at 4–6 kW (Table IV). These fixed-implementation counts do not imply topology-level superiority.

TABLE IV
MODELED-EFFICIENCY LEADS BY POWER

Power	DAB	CLLC	Numeric coverage	Interpretation
1 kW	1	9	10/10	CLLC higher at 9 of 10 points
2 kW	0	10	10/10	CLLC higher at all 10 points
3 kW	2	8	10/10	CLLC higher at 8 of 10 points
4 kW	7	3	10/10	DAB higher at 7 of 10 points
5 kW	9	1	10/10	DAB higher at 9 of 10 points
6 kW	10	0	10/10	DAB higher at all 10 points
Total	**29**	**31**	**60/60**	Load- and voltage-ratio-dependent crossover

Fig. 5 relates the observed current and efficiency differences to the voltage ratio $d = nV_{bat}/V_{dc}$. At 1 kW, the CLLC-to-DAB primary-rms-current ratio is lowest where SPS-DAB voltage mismatch is largest and approaches or exceeds unity near $d = 1$. The corresponding efficiency bars show that the light-load advantage changes with battery voltage, whereas the 6-kW baseline modeled ordering favors DAB. At 6 kW, the predefined CLLC duty reaches unity, so the fundamental-voltage matching commanded at 1 kW is not retained. This association is consistent with a voltage-mismatch penalty under the retained SPS control and the CLLC fundamental-voltage schedule; it is not proof that either mechanism is the sole cause. Extended- or dual-phase-shift DAB control [6] was not evaluated. Across the matrix, primary rms/peak currents reach 25.33/39.13 A for DAB and 28.59/39.20 A for CLLC. At G2V 300 V/6 kW, conduction/switching losses are 90.46/17.98 W for DAB and 94.13/15.79 W for CLLC; lower switching loss alone does not determine total efficiency.

IV. Sensitivity and Robustness Assessment

A. Exact-Power Near-Tie Validation

Among 17 preselected near-ties, 15 have qualified independent exact-power confirmations at 2 ns (Table V). Both converters must meet the Section II-B numerical gates and a stricter 0.05% target-power tolerance. Phase-only searches confirmed G2V 300 V/3 kW and 450 V/4 kW. All 15 confirmations preserve the baseline ordering. Two remain unresolved. These checks update evidence states, not the nominal 60-pair values.

B. Timestep Sensitivity

All 22 timestep-check pairs were preselected. Bilateral 0.5-ns checks resolved three of seven open pairs; predefined 0.25-ns checks resolved the other four. All 22 qualify at the tested resolutions and preserve the baseline Δη sign, demonstrating stability only in this subset and resolution range. Two V2G 250-V pairs (1 and 4 kW) move from additional validation to warning; the nominal 31/29 ordering is unchanged.

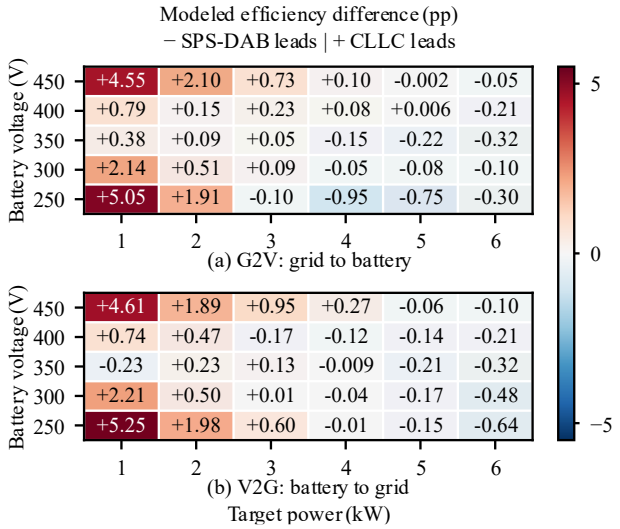


Fig. 3. Modeled-efficiency difference $\Delta\eta = \eta_{CLLC} - \eta_{DAB}$ for all 60 pairs: (a) G2V and (b) V2G. Each cell reports the signed value in percentage points; positive (red) favors CLLC and negative (blue) favors DAB.

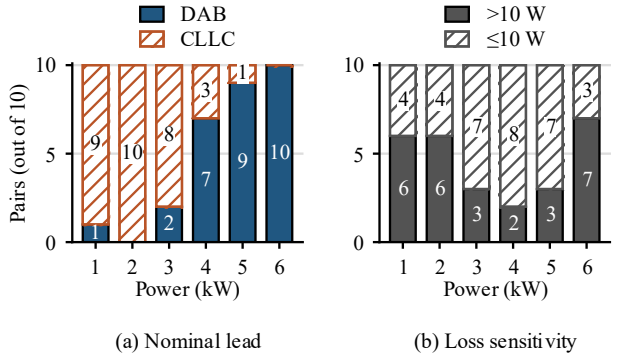


Fig. 4. Ten voltage–direction pairs per power level: (a) nominal modeled-efficiency lead and (b) count above or at/below the 10-W differential omitted-loss threshold. The threshold is a sensitivity calculation, not measured missing loss.

C. Omitted-Loss Robustness

The omitted-loss assessment adds differential loss to the nominally leading implementation until its efficiency equals that of the other implementation, as defined in (4). Fig. 6 shows the distribution: five thresholds are below 1 W, seven lie from 1 to <3 W, ten from 3 to <5 W, eleven from 5 to ≤10 W, and 27 exceed 10 W. Thus, 33 of 60 orderings are sensitive at or below 10 W, which corresponds to 1% of the received power at 1 kW and about 0.17% at 6 kW; 18 of these are DAB leads and 15 are CLLC leads. The threshold indicates ranking sensitivity; it is not an estimate of physically missing loss.

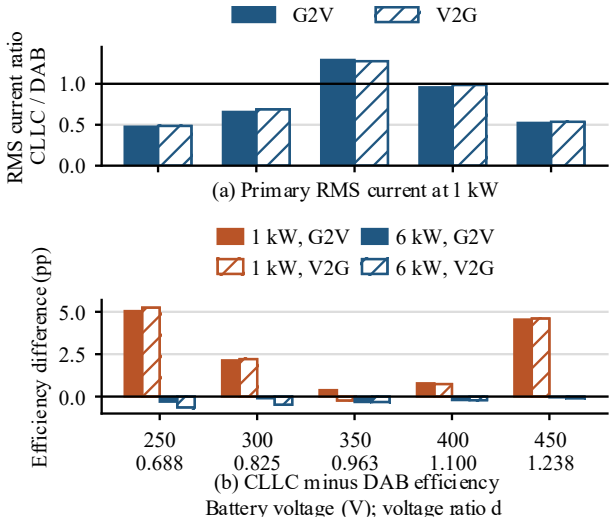


Fig. 5. Voltage-ratio interpretation: (a) CLLC-to-DAB primary-rms-current ratio at 1 kW and (b) modeled-efficiency difference at 1 and 6 kW.

D. Soft-Switching, Temperature, and Model Fidelity

Fig. 7 compares steady-state waveforms at G2V 400 V/4 kW. The SPS-DAB primary current is piecewise linear, whereas the CLLC current is quasi-sinusoidal at 100 kHz, 1.25 times the branch resonance. Across the 120 records, 49,152 commutation events include 24,576 selected turn-on events. The turn-on proxy passes 6,280 of 9,600 DAB events and 11,700 of 14,976 CLLC events; 38 operating-point pairs retain soft-switching-margin warnings. These event-weighted fractions describe the polarity, voltage-clamp, charge, and energy screen; they are not physical ZVS success rates. The screen's energy and charge surrogates do not integrate nonlinear Q_{oss} or reproduce measured device commutation.

Recomputing device losses at 25, 125, and 175 °C from the selected baseline waveforms gives 55 stable and five temperature-sensitive pairs. The 125 °C components reproduce the baseline to numerical precision; V2G 350 V/1 kW is therefore reclassified from additional validation to warning. Because no coupled thermal network or measured junction temperature is present, the result is a parameter sensitivity rather than electrothermal validation.

Bridge-model fidelity. DAB bridges use discrete MOSFETs whereas CLLC bridges are parameterized (Table II). Six fixed-control CLLC points were rebuilt with explicit MOSFETs and common loss post-processing. Interpretability required per-device off-gate reverse conduction ≤0.5 µs/cycle, paired-switch loss difference ≤10%, gate-aligned channel charge and sample coverage ≥90%, and Section II-B numerical qualification. Three points passed. V2G 350 V/4 kW failed the 0.5% power gate (error 0.616%), and two failed other numerical gates. At G2V 300 V/3 kW, 450 V/5 kW, and 350 V/6 kW, included CLLC loss decreased by 3.37–13.58 W (Δη shift +0.11–0.22 pp at baseline received power). The 450-V/5-kW near tie reversed (threshold 0.10 W); the 350-V/6-kW DAB lead survived a 13.58-W change below its 20.1-W threshold. A separate phase-only pilot qualified one of three conditions at 1 ns and 0.05% power error; two lacked confirmation owing to resource limits. These are local sensitivity results, not a map-wide bias estimate. Phase retuning also changes current, and neither cohort replaces the baseline map.

E. *Independent Analytical Cross-Check*

Without further simulation, the switched records were compared with first-principles models (Fig. 8). For the DAB, the ideal SPS inductor current was reconstructed from (1) at each record's transferred power, approximated as P_{rec} plus half of $P_{loss,inc}$. The reconstructed primary rms current agrees with the PSIM value within 1.8% at all 60 points and within 3.7% for any split of the included loss between the two bridges. For the CLLC, the lossless first-harmonic approximation (FHA) T-network gives

$$P_t = [\hat{V}_{s1}\,\hat{V}_{r1}/(2X_T)]\sin\theta \qquad (5)$$

with primary-referred reactances $X_T = X_1 + X_2 + X_1X_2/X_m$, $X_1 = \omega L_{r1} - 1/(\omega C_{r1})$, $X_2 = n^2[\omega L_{r2} - 1/(\omega C_{r2})]$, $X_m = \omega L_m$, and $\omega = 2\pi f_s$. For DC amplitude V and pulse fraction D, the commanded fundamental peak is $\hat{V}_1 = (4V/\pi)\sin(\pi D/2)$. Using $P_t \approx P_{rec} + P_{loss,inc}/2$, (5) predicts the principal-branch electrical phase θ between bridge-voltage fundamentals.

Direct Fourier integration and a discrete Fourier transform applied to the same native bridge-voltage waveforms yield consistent estimates of their fundamental electrical phase. The FHA prediction is within ±5% at 58 of 60 points; the G2V 250-V/1-kW and 300-V/1-kW residuals are −8.32% and −5.94%. Gate-command phase differs from the measured fundamental phase by up to 7.07° and is retained as a separate diagnostic. These comparisons support analytical consistency, not switching-loss, soft-switching, or hardware-efficiency validation.

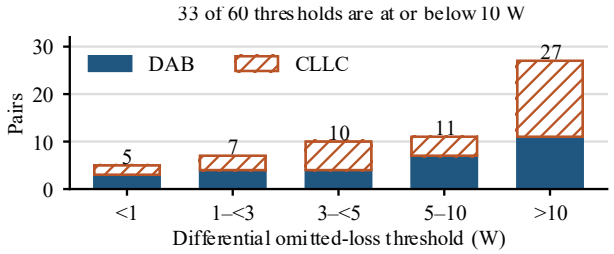


Fig. 6. Distribution of differential omitted-loss reversal thresholds from (4), grouped by nominal lead. The five bin counts sum to 60; 33 thresholds are at or below 10 W and 27 exceed 10 W.

TABLE V
SENSITIVITY ASSESSMENT SUMMARY

Assessment	Result	Interpretation limit
Exact-power alignment	15 of 17 eligible; 2 unresolved	Separate 2-ns test; 0.05% power tolerance
Timestep sensitivity	22 of 22 stable; none unresolved	Stable within tested 1/0.5/0.25-ns resolutions
Unmodeled-loss sensitivity	33 of 60 thresholds ≤10 W	Ranking sensitivity, not measured missing loss
Temperature sensitivity	55 stable; 5 sensitive	Device-parameter sensitivity only
Soft-switching diagnostics	120 records; 49,152 events; 38 pair warnings	Model-defined event fractions; no physical ZVS/ZCS proof
Bridge-model fidelity	Original: 3/6; phase-only follow-up: 1/3	Local static-model checks; no full-map bound
Analytical cross-check	DAB rms ≤1.8% (60/60); CLLC fundamental phase ±5% (58/60)	Model consistency only; not loss or ZVS validation

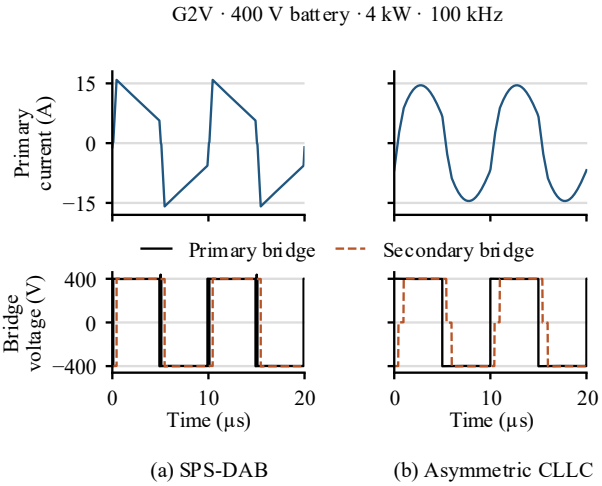


Fig. 7. Native PSIM waveforms at G2V, 400-V battery, and 4 kW: (a) SPS-DAB and (b) asymmetric CLLC. Top: primary current; bottom: primary and secondary bridge voltages. Both panels show the same two 100-kHz cycles from the 1-ns records, without smoothing or resampling.

V. Discussion and Design Implications

A. *Load- and Voltage-Ratio-Dependent Trends*

Of the 12 pairs supported by the model-screening criteria, ten nominally favor DAB and two CLLC. Evidence states reflect numerical, sensitivity, and soft-switching criteria, not the magnitude of Δη or a physical omitted-loss bound. Thus, large CLLC margins may retain warnings while smaller DAB leads pass the baseline screen. Voltage-ratio dependence remains clear at 1 kW (Fig. 5).

B. *Fairness and Model Limitations*

The common operating matrix does not equalize optimization opportunity. Unequal bridge-model fidelity remains a key uncertainty; explicit-MOSFET results apply locally and cannot correct the whole map. No physical bound on omitted inductor losses is established. Gap/fringing, thermal and gate-loop parasitics, auxiliaries, protection, electromagnetic compatibility, reliability, and manufacturability remain outside scope.

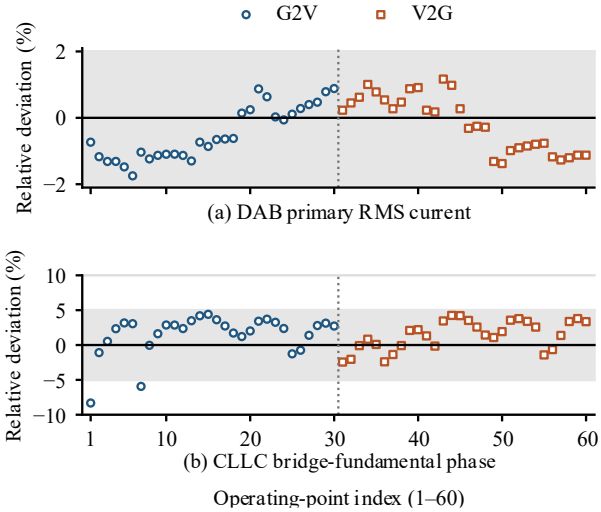


Fig. 8. Relative deviations between simulation and ideal analytical approximations at all 60 points: (a) PSIM relative to SPS rms current; (b) FHA relative to the bridge-voltage fundamental phase extracted from native PSIM samples. Bands: ±2% in (a), ±5% in (b). Indices 1–30: G2V; 31–60: V2G, ordered by voltage then power.

C. Implications for Converter Selection

The modeled Δη map is interpreted with current, losses, and evidence state. (i) At 1–2 kW and 250, 300, or 450 V, CLLC nominally leads all 12 pairs by 0.50–5.25 pp; 11 remain below the supported evidence tier, limiting physical confidence but not changing the reported margins. (ii) At 350–400 V and 1–3 kW, $|\Delta\eta| \leq 0.79$ pp; current and switching stress matter alongside efficiency. (iii) At 4–6 kW, DAB nominally leads 26/30. Of those 26, 19 have margins below 0.25 pp, comparable to the +0.11–0.22-pp local explicit-MOSFET CLLC shifts. These shifts are not valid global corrections. Ten DAB leads are supported (0.21–0.95 pp); G2V 350 V/6 kW retained its lead when locally retested. At G2V 300 V, included losses are 26.26 W (CLLC) versus 49.25 W (DAB) at 1 kW, versus 144.86 W (CLLC) and 137.97 W (DAB) at 6 kW. All rules are implementation-specific screens, not hardware recommendations.

VI. Conclusion

Predefined SPS-DAB and asymmetric CLLC implementations were compared across a 400-V link, 250–450-V battery range, 1–6 kW, and both power directions. CLLC nominally leads 31 pairs, chiefly at light load and SPS voltage mismatch; DAB nominally leads 29, including 26/30 at 4–6 kW. Nineteen of those 26 high-load DAB margins are below 0.25 pp, comparable in scale to local explicit-MOSFET CLLC sensitivity results. No map-wide model-bias correction is justified. Screening assigns 12 pairs to supported, 44 to warning, and four to additional validation; 33 have hypothetical differential omitted-loss reversal thresholds ≤10 W. These findings guide screening of the evaluated fixed implementations, not universal or hardware-validated topology selection. Same-fidelity map tests, flexible DAB modulation, and hardware loss/ZVS measurements remain future work. Underlying records, checksums, and scripts are available from the first author upon reasonable request.

References

[1] M. Khanra, M. Singh, and M. Klobasa, "Vehicle-to-Grid for Thailand: A global V2G best practices," GIZ, Rep. 1/V2G, TGC-EMC, Karlsruhe, Germany, 2025. Accessed: Nov. 4, 2026. [Online]. Available: https://www.thai-german-cooperation.info/resource/vehicle-to-grid-for-thailand-a-global-v2g-best-practices/

[2] P. He and A. Khaligh, "Comprehensive analyses and comparison of 1 kW isolated DC-DC converters for bidirectional EV charging systems," IEEE Trans. Transp. Electrific., vol. 3, no. 1, pp. 147–156, Mar. 2017, doi: 10.1109/TTE.2016.2630927.

[3] K. Siebke and R. Mallwitz, "Comparison of a dual active bridge and CLLC converter for on-board vehicle chargers using GaN and time domain modeling method," in Proc. IEEE Energy Convers. Congr. Expo. (ECCE), 2020, pp. 1210–1216, doi: 10.1109/ECCE44975.2020.9236260.

[4] H.-S. Oh, S.-Y. Hong, J. Lee, and J.-B. Lee, "Comparison of bi-directional topologies for on-board charger: A 10.9 kW high-efficiency high power density of DC-DC stage," Energies, vol. 17, no. 21, Art. no. 5496, Nov. 2024, doi: 10.3390/en17215496.

[5] M. S. Chowdhary, F. I. Bakhsh, M. Amir, A. F. Minai, F. Ahmad, and M. Z. Ansari, "Comprehensive analysis and comparison of DAB and CLLC converters for EV application," in Proc. IEEE 4th Int. Conf. Sustainable Energy Future Electr. Transp. (SEFET), 2024, pp. 1–6, doi: 10.1109/SEFET61574.2024.10717946.

[6] N. Hou and Y. W. Li, "Overview and comparison of modulation and control strategies for a nonresonant single-phase dual-active-bridge DC-DC converter," IEEE Trans. Power Electron., vol. 35, no. 3, pp. 3148–3172, Mar. 2020, doi: 10.1109/TPEL.2019.2927930.

[7] Y. Park, S. Chakraborty, and A. Khaligh, "DAB converter for EV onboard chargers using bare-die SiC MOSFETs and leakage-integrated planar transformer," IEEE Trans. Transp. Electrific., vol. 8, no. 1, pp. 209–224, Mar. 2022, doi: 10.1109/TTE.2021.3121172.

[8] J. Min and M. Ordonez, "Bidirectional resonant CLLC charger for wide battery voltage range: Asymmetric parameters methodology," IEEE Trans. Power Electron., vol. 36, no. 6, pp. 6662–6673, Jun. 2021, doi: 10.1109/TPEL.2020.3033982.

[9] Z. U. Zahid, Z. M. Dalala, R. Chen, B. Chen, and J.-S. Lai, "Design of bidirectional DC–DC resonant converter for vehicle-to-grid (V2G) applications," IEEE Trans. Transp. Electrific., vol. 1, no. 3, pp. 232–244, Oct. 2015, doi: 10.1109/TTE.2015.2476035.

[10] L. Farias Martins, D. A. Stone, and M. P. Foster, "State-variable and cyclic-averaging analysis of bidirectional CLLC resonant converters," J. Eng., vol. 2019, no. 17, pp. 4364–4368, 2019, doi: 10.1049/joe.2018.8072.

[11] P. Jia, Y. Zhang, M. Liu, and Y. Xing, "Model analysis and design of the control strategy for the bidirectional resonant DAB converter," IEEE Trans. Power Electron., vol. 40, no. 7, pp. 9556–9576, Jul. 2025, doi: 10.1109/TPEL.2025.3543718.

[12] B. Oude Aarninkhof, D. Lyu, T. B. Soeiro, and P. Bauer, "A reconfigurable two-stage 11 kW DC-DC resonant converter for EV charging with a 150–1000 V output voltage range," IEEE Trans. Transp. Electrific., vol. 10, no. 1, pp. 509–522, Mar. 2024, doi: 10.1109/TTE.2023.3279211.

[13] A. Elezab, O. Zayed, A. Abuelnaga, and M. Narimani, "High efficiency LLC resonant converter with wide output range of 200–1000 V for DC-connected EVs ultra-fast charging stations," IEEE Access, vol. 11, pp. 33037–33048, 2023, doi: 10.1109/ACCESS.2023.3263486.

[14] L. Li, Y. Cao, L. Hou, T. Mao, and W. Ning, "PFM-PS hybrid control strategy CLLC resonant converter research," in Proc. 8th Asia Conf. Power Electr. Eng. (ACPEE), 2023, pp. 1278–1284, doi: 10.1109/ACPEE56931.2023.10135949.

[15] S. Zou, J. Lu, A. Mallik, and A. Khaligh, "Bi-directional CLLC converter with synchronous rectification for plug-in electric vehicles," IEEE Trans. Ind. Appl., vol. 54, no. 2, pp. 998–1005, Mar./Apr. 2018, doi: 10.1109/TIA.2017.2773430.

[16] Q. Zhao, Y. Gao, H. Ding, Z. Wu, X. Li, and D. Wang, "Bidirectional hybrid DC–DC resonant converter with wide voltage gain range," IEEE Trans. Ind. Electron., vol. 72, no. 5, pp. 4721–4730, May 2025, doi: 10.1109/TIE.2024.3468726.

[17] H. Li et al., "A multiobjective optimized asymmetric triple-variable modulation strategy and simplified direct duty scheme for DAB converters," IEEE Trans. Power Electron., vol. 41, no. 4, pp. 6290–6303, Apr. 2026, doi: 10.1109/TPEL.2025.3628062.

[18] P. He, A. Mallik, A. Sankar, and A. Khaligh, "Design of a 1-MHz high-efficiency high-power-density bidirectional GaN-based CLLC converter for electric vehicles," IEEE Trans. Veh. Technol., vol. 68, no. 1, pp. 213–223, Jan. 2019, doi: 10.1109/TVT.2018.2881276.

[19] J.-W. Yeom, S.-M. Jeong, C.-M. Lim, and I.-O. Lee, "Design and experimental validation of a 25-kW dual DAB–CLLC bidirectional DC/DC converter with an ultrawide 200–1000 V operating range," IEEE Access, vol. 14, pp. 63626–63642, 2026, doi: 10.1109/ACCESS.2026.3686835.

[20] R. P. Upputuri and B. Subudhi, "A comprehensive review and performance evaluation of bidirectional charger topologies for V2G/G2V operations in EV applications," IEEE Trans. Transp. Electrific., vol. 10, no. 1, pp. 583–595, Mar. 2024, doi: 10.1109/TTE.2023.3289965.

[21] A. Phimphui and U. Supatti, "V2G and G2V using interleaved converter for a single-phase onboard bidirectional charger," in Proc. 2019 IEEE Transp. Electrific. Conf. Expo, Asia-Pacific (ITEC Asia-Pacific), 2019, pp. 1–5, doi: 10.1109/ITEC-AP.2019.8903662.

[22] TDK Electronics, "E 65/32/27 core and accessories, series B66387/B66388," data sheet, Oct. 2024. Accessed: Nov. 4, 2026. [Online]. Available: https://www.tdk-electronics.tdk.com/inf/80/db/fer/e_65_32_27.pdf

[23] Wolfspeed, "C3M0032120K silicon carbide power MOSFET," data sheet, Rev. 6, Sep. 2024. Accessed: Nov. 4, 2026. [Online]. Available: https://assets.wolfspeed.com/uploads/2024/01/Wolfspeed_C3M0032120K_data_sheet.pdf

[24] Altair Engineering, "1-ph Inverter," PSIM 2024.1 Help, SM_1PH_INV, ideal-model parameters, installed documentation, accessed Sep. 27, 2026.